ALA - Assignment -3(a)

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Q1.

a) A classifier has K = 5 classes. e_x - whose true class is class 2, write down both label c and its target y. What does column j of a weight matrix $W^{(l)}$ contain

→ label is a number, index of true class.
target is a vector of length K with single 1 at the index.

so label ⇒ c = 2 & Target ⇒ $y = (0,1,0,0,0)^T$.

→ Column j of $W^{(l)}$ contains all weights connecting unit j of the previous layer (l-1) to every unit of Layer l..

b) A fully connected network has layer sizes 4 → 3 → 2.
Give the shape of each weight matrix & bias vector, count the weights, the biases and the total no. of parameters.

→ $W^{(l)}$ has 1 row per unit of layer l & 1 column per unit layer l-1,
so its shape is $n_l \times n_{l-1}$

$b^{(l)}$ bias has one entry per unit of layer l.

layer 1 (4 inputs ; 3 units) : $W^{(1)} \in \mathbb{R}^{3\times 4}$, so 12 weights & $b^{(1)} \in \mathbb{R}^3$, so 3 biases

layer 2 (3 inputs ; 2 units : $W^{(2)} \in \mathbb{R}^{2\times 3}$, so 6 weights & $b^{(2)} \in \mathbb{R}^2$, so 2 biases

total = 12+6 = 18 weights & 3+2 = 5 biases, so 23 parameters

c) Explain in your words, why Section 2 takes logarithm of the likelihood, and why it then negates it. For the target $y = (0,1,0,0,0)^T$ show that $\prod_{k=1}^{5} q_k^{y_k}$ is equal to q_2

→ likelihood is the product of probabilities of the observed correct classes.
Multiplying many small probabilities can cause numerical underflow..

taking logarithms converts multiplication into addition :-

$\ln(ab) = \ln a + \ln b$

it also make differentiation easier. since logarithms is an increasing function maximizing likelihood is equivalent to minimizing log-likelihood.

Why negate?

↳ Optimizers are written to minimize. Maximizing ℓ is same as minimizing $-\ell$, the negative log-likelihood.

the product, write out all factors:

$$\prod_{k=1}^{5} q_k^{y_k} = q_1^0 \, q_2^1 \, q_3^0 \, q_4^0 \, q_5^0 = 1 \cdot q_2 \cdot 1 \cdot 1 \cdot 1 = q_2$$

every class with $y_k = 0$ contributes a factor 1, so only the true class is left.

d) A freshly initialized network with small random weights is trained on $K = 5$ classes. About what loss should it give on its first example & why?

→ small weights give logits that are all close to zero, so the softmax cannot prefer any class.
with all logits nearly equal, every $q_k \approx \frac{1}{K} = \frac{1}{5}$
the loss is $L = -\ln q_c \approx -\ln(1/5) = \ln(5) \approx 1.6094$ nats
whatever the labels. A much larger first loss usually means a bug or weights that are too large.

e) For a model output $q = (0.05, 0.7, 0.1, 0.1, 0.05)^T$, $c = 2$

Compute $\delta^{(3)} = q - y$ & check that its entries sum to zero.

Which entry is negative, and what does that tell you about how a gradient-descent step changes the logits.

Section (2) shows that the gradient at the logits is simply the prediction minus the target.

$\Rightarrow \therefore \frac{\partial L}{\partial z_k} = q_k - y_k$ so $\delta^{(3)} = q - y$.

$c = 2$

$y = (0, 1, 0, 0, 0)^T$

$\delta^{(3)} = q - y = (0.05 - 0, 0.7 - 1, 0.1 - 0, 0.1 - 0, 0.05 - 0)^T$

$= (0.05, -0.3, 0.1, 0.1, 0.05)^T$

sum → 0.05 + (−0.3) + (0.1) + 0.1 + 0.05 . = 0

= 0

This slightly happens because $\sum_k q_k = 1$ and $\sum_k y_k = 1$, so

$\sum_k (q_k - y_k) \cdot -1 = 0$. only the true class has negative sign

→ the gradient pushes its logit up and the others down.

f) A layer has 3 inputs a_1, a_2, a_3 & 2 units with

$z_i = w_{i1} a_1 + w_{i2} a_2 + w_{i3} a_3 + b_i$ for $i = 1, 2$ & $\delta_i = \frac{\partial L}{\partial z_i}$ is known

Write out z_1 & z_2. Find the $\partial L / \partial w_{13}$ & $\partial L / \partial b_1$. Arrange the 6 weight gradients in a matrix. and shows that it equals the outer product of δ & a.

→ In section 4.1, I'll work out each z_i & see which weights it contains.

A weight appears in only 1 z_i, so only one term of the chain rule survives.

$z_1 = w_{11} a_1 + w_{12} a_2 + w_{13} a_3 + b_1$ w_{13} appears in z_1.

$z_2 = w_{21} a_1 + w_{22} a_2 + w_{23} a_3 + b_2$

~~$z_3 = w_{31} a_1 + w_{32} a_2 + w_{33} a_3 + b_3$~~

w_{13} appears only in z_1, multiplied by a_3 so $\frac{\partial z_1}{\partial w_{13}} = a_3$

& $\frac{\partial L}{\partial w_{13}} = \frac{\partial L}{\partial z_1} \frac{\partial z_1}{\partial w_{13}} = \delta_1 a_3$

b_1 appears only in z_1, as a plain added term, so $\frac{\partial z_1}{\partial b_1} = 1$ &

$\partial L / \partial b_1 = \delta_1$ in same way $\partial L / \partial w_{ij} = \delta_i a_j$ — write this

for every i, j. Placing entry (i, j) in row i & column j:

we obtain $\frac{\partial L}{w_{11}} = \delta_1 a_1$ $\frac{\partial L}{w_{21}} = \delta_2 a_1$ $\frac{\partial L}{} = \delta_3 a_1$

$\frac{\partial L}{w_{12}} = \delta_1 a_2$ $\frac{\partial L}{w_{22}} = \delta_2 a_2$ $\frac{\partial L}{} = \delta_3 a_2$

$\frac{\partial L}{w_{13}} = \delta_1 a_3$ $\frac{\partial L}{w_{23}} = \delta_2 a_3$ $\frac{\partial L}{w_{33}} = \delta_3 a_3$

$\frac{\partial L}{} = \begin{bmatrix} \delta_1 a_1 & \delta_1 a_2 & \delta_1 a_3 \\ \delta_2 a_1 & \delta_2 a_2 & \delta_2 a_3 \\ \delta_3 a_1 & \delta_3 a_2 & \delta_3 a_3 \end{bmatrix}$

matrix $\begin{pmatrix} \delta_1 a_1 & \delta_1 a_2 & \delta_1 a_3 \\ \delta_2 a_1 & \delta_2 a_2 & \delta_2 a_3 \end{pmatrix} = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} [a_1 \; a_2 \; a_3] =$

$= \delta a^T$

Row i is δ_i times the input vector; column j is a_j times δ.

Show it works to outer product.

let $\delta = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix}$ & $a^T = [a_1 \; a_2 \; a_3]$

$\delta a^T = \begin{bmatrix} \delta_1 \\ \delta_2 \end{bmatrix} [a_1 \; a_2 \; a_3] = \begin{bmatrix} \delta_1 a_1 & \delta_1 a_2 & \delta_1 a_3 \\ \delta_2 a_1 & \delta_2 a_2 & \delta_2 a_3 \end{bmatrix}$

$\Rightarrow$ exactly same as weight gradient matrix.

MA - Assignment - 3(b)

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Q1

$$W^{(1)} = \begin{pmatrix} 0.4 & -0.2 \\ -0.5 & 0.3 \\ 0.2 & 0.6 \end{pmatrix}, \quad b^{(1)} = \begin{pmatrix} 0 \\ 0.1 \\ 0 \end{pmatrix}$$

$$W^{(2)} = \begin{pmatrix} 0.3 & 0.5 & -0.2 \\ -0.4 & 0.2 & 0.5 \\ 0.6 & -0.3 & 0.1 \end{pmatrix} \quad W^{(3)} \begin{pmatrix} 0.2 & 0.4 & -0.3 \\ 0.5 & -0.1 & 0.2 \\ -0.2 & 0.3 & 0.4 \end{pmatrix}$$

$b^{(2)} = b^{(3)} = 0$. input $x = (2,1)^T$ & label $c = 3$.

a) Compute $z^{(1)} = W^{(1)}x + b^{(1)}$ and $a^{(1)} = ReLU(z^{(1)})$. Which unit of layer 1 is inactive? Does the bias change which units are active here?

Soln $W^{(1)} = \begin{bmatrix} 0.4 & -0.2 \\ -0.5 & 0.3 \\ 0.2 & 0.6 \end{bmatrix}$ $b^{(1)} = \begin{bmatrix} 0 \\ 0.1 \\ 0 \end{bmatrix}$ $x = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.

Req: $z^{(1)}$, $a^{(1)}$, identify the inactive unit and determine whether bias changes the active units.

formula = $z^{(1)} = W^{(1)}x + b^{(1)}$

$\therefore ReLU(z) = \max(0, z)$

(1) calculate the pre-activation vector.

$$z^{(1)} = \begin{bmatrix} 0.4 & -0.2 \\ -0.5 & 0.3 \\ 0.2 & 0.6 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0.1 \\ 0 \end{bmatrix}$$

$$z^{(1)} = \begin{bmatrix} 0.6 \\ -0.6 \\ 1.0 \end{bmatrix}$$

(2) Apply ReLU activation.

$$a^{(1)} = ReLU(z^{(1)}) = \begin{bmatrix} \max(0, 0.6) \\ \max(0, -0.6) \\ \max(0, 1.0) \end{bmatrix} \Rightarrow a^{(1)} = \begin{bmatrix} 0.6 \\ 0 \\ 1.0 \end{bmatrix}$$

inactive unit

The 2nd unit has: $z_2^{(1)} = -0.6$

After ReLU = $a_2^{(1)} = 0$.

$\therefore$ unit 2 is inactive

effect of bias

without bias $= z^{(1)} = \begin{bmatrix} 0.6 \\ -0.2 \\ 1.0 \end{bmatrix}$ | with bias $= z^{(1)} = \begin{bmatrix} 0.6 \\ 0.6 \\ 1.0 \end{bmatrix}$.

in both cases units 1, 3 are active unit a^2 is inactive
Bias change the value of second unit but does not change which units are active

b) Compute $z^{(2)}$ & $a^{(2)}$ which column of $W^{(2)}$ has no effect on $z^{(2)}$? Why?

Sol- $a^{(1)} = \begin{bmatrix} 0.6 \\ 0 \\ 1.0 \end{bmatrix}$ $\quad b^{(2)} = 0$.

$$W^{(2)} = \begin{bmatrix} 0.3 & 0.5 & -0.2 \\ -0.4 & 0.2 & 0.5 \\ 0.6 & -0.3 & 0.1 \end{bmatrix}$$

$$z^{(2)} = W^{(2)} a^{(1)} + b^{(2)}$$

$$a^{(2)} = \text{ReLU}(z^{(2)})$$

calculate $z^{(2)}$

$$= \begin{bmatrix} -0.3 & 0.5 & -0.2 \\ -0.4 & 0.2 & 0.5 \\ 0.6 & -0.3 & 0.1 \end{bmatrix} \begin{bmatrix} 0.6 \\ 0 \\ 1.0 \end{bmatrix}$$

row wise multiplication
After calculation we get :- $z^{(2)} = \begin{bmatrix} -0.02 \\ 0.26 \\ 0.46 \end{bmatrix}$

ReLU $a^{(2)} =$ [illegible] $\begin{bmatrix} \max(0, -0.02) \\ \max(0, 0.26) \\ \max(0, 0.46) \end{bmatrix} = \begin{bmatrix} 0 \\ 0.26 \\ 0.46 \end{bmatrix}$

Columns having no effect the 1st layer output is! -

$$a^{(1)} = \begin{bmatrix} 0.6 \\ 0 \\ 1.0 \end{bmatrix}$$

Since its 2nd second component is zero the second column of $W^{(2)}$ contributes nothing.

The 2nd column of $W^{(4)}$ has no effect because the (corresponding) input activation is zero.

c) Compute the logits $z^{(3)}$. which column $W^{(3)}$ has not effect hence?

Soln :- $W^{(3)} = \begin{bmatrix} 0.2 & 0.4 & -0.3 \\ 0.5 & -0.1 & 0.2 \\ -0.2 & 0.3 & 0.4 \end{bmatrix}$

$$a^{(2)} = \begin{bmatrix} 0 \\ 0.26 \\ 0.46 \end{bmatrix} \quad b^{(3)} = 0$$

$$z^{(3)} = W^{(3)} a^{(2)} + b^{(3)}$$

$$= \begin{bmatrix} 0.2 & 0.4 & -0.3 \\ 0.5 & -0.1 & 0.2 \\ -0.2 & 0.3 & 0.4 \end{bmatrix} \begin{bmatrix} 0 \\ 0.26 \\ 0.46 \end{bmatrix}$$

calculation row by row wise

$$z^{(3)} = \begin{bmatrix} -0.34 \\ 0.066 \\ 0.262 \end{bmatrix}$$

Columns having zero effect

→ 1st component of $a^{(2)}$ is zero

→ the 1st column of $W^{(3)}$ has no effect on logits.

d) Compute q = softmax $(z^{(3)})$ and check that its entries add to 1

Compute loss $L = -\ln q_c$. Is L above or below $\ln 3 \approx 1.0986$ & why?

Soln $z^{(3)} = \begin{bmatrix} -0.034 \\ 0.066 \\ 0.262 \end{bmatrix}$ $c = 3 \rightarrow$ true class

Softmax $= q_k = \dfrac{e^{z_k}}{\sum_{j=1}^{3} e^{z_j}}$

cross entropy loss $L = -\ln q_c$.

Calculate exponential values :-

$e^{-0.034} = 0.9666$

$e^{0.066} = 1.0682$

$e^{0.262} = 1.2995$.

Sum exponential value =

$S \approx 3.3343$

Softmax probability

For class 1: $q_1 = \frac{0.9666}{3.3343}$

$q_1 = 0.2899$

For class 2: $q_2 = \frac{1.0682}{3.3343}$

$q_2 = 0.3204$

For class 3: $q_3 = \frac{1.2995}{3.3343}$

$= 0.3897$

$$q = \begin{bmatrix} 0.2899 \\ 0.3204 \\ 0.3897 \end{bmatrix}$$

Verify that probability sum to 1.

$$\sum_{k=1}^{3} q_k = q_1 + q_2 + q_3$$

$$= 0.2899 + 0.3204 + 0.3897$$

$$= 1$$

The probabilities are correctly normalized.

Calculate cross entropy loss.

Given that true class is class 3.

$$L = -\ln q_3$$

$$L = -\ln(0.3897)$$

$$L \approx 0.9423$$

Compare with $\ln 3$

$\ln 3 \approx 1.0986$

We have

$0.9423 < 1.0986$.

Also $q_3 = 0.3897 > 1/3$

∴ loss is below $\ln 3$ because the model assigns a probability greater than the uniform probability to the correct class.